DLD Project Proposal

[Mujtaba Hasan, Laiba Ahmed, Zain Usmani, Sumaira Khan]

Project Name:

چھیاسٹھ کتنے ہوتے ہیں؟

Introduction:

How many times has it happened that someone says something in urdu and we don’t understand it? (Very often, we know). The vast majority of people don’t know the full urdu alphabet or even how to count to 20 in urdu and that's very worrying. And hence, this is why we are creating this project. Our project will not even teach people the urdu numerals(Or alphabets) but also test them on how much they’ve learned.

Description:

Our product will rely on two modules/modes:

1) Testing mode
2) Learning mode*

Testing Mode:

In this mode the user prompts the device to output the audio of a random number(or alphabet) in urdu. The user will listen to the audio and recognize what the number or alphabet it is. After the user recognizes what it is, they input that number or alphabet to the device and it checks if the user is right or not.

The input from the user can be taken in three different ways.

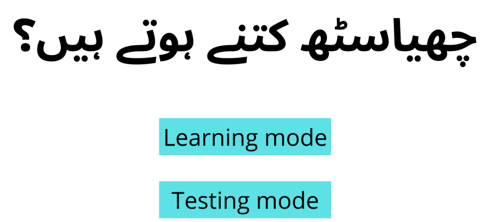
1) **Via a keyboard:** In response to the audio, the user presses the corresponding key. For example the device says پانچ , the user will press the “5” key or if the device says ک they press the corresponding key that is marked as ک on the keyboard.
2) **Via a mouse:** In response to the audio, the user can draw the number or alphabet onto the screen. We will limit the area in which the user can “draw” .This will be done by using a 23x23 pixel display. The error margin will be adjusted after testing.
3) **Via a touchpad:** In response to the audio, the user can draw the number or alphabet with their finger onto a touchpad. We will be using multiple sensors for this and as with the mouse input the error margin will be adjusted after testing.

Since our design is still in its planning mode we have yet to decide which mode of input to go with.

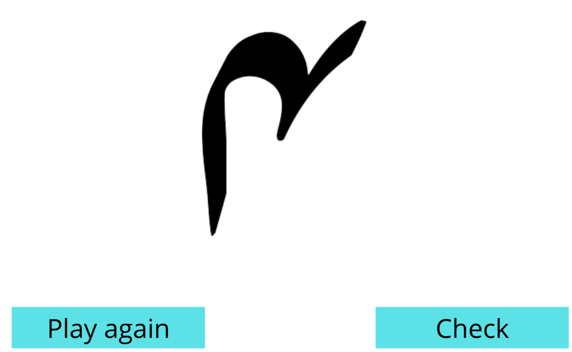
Learning Mode*:
The learning mode displays numbers(or alphabets) onto the screen. The user can select any number (or alphabet) and its audio pronunciation will play. The input from the user can be taken in two ways:

1) **Via keyboard:** The user can press a key say 7 on the keyboard and in response the audio pronunciation for سات will play.
2) **Via mouse:** The user can click on any number displayed on the VGA and the audio pronunciation will play.

Prototype:
Our project will have potentially two or three different displays. There will be an initial display from which the user can decide which module they want to go to (learning or testing).



From here the user can go to either of the two modules. If the user decides to go to the testing module they would a display similar to:



Further Notes:

1) Our priority will be to complete the testing module since that uses more dld concepts. We will only add the learning module once our testing module is complete.
2) We will increase the number of urdu numerals our device can deal with once we start prototyping our model on hardware.
3) We can either go with urdu numbers or alphabets for our projects. The functionality of our project will be the same but we need to decide what we're going to use in our final design.